

- (c) The restaurant offers a discount on a meal if the person ordering is either a student or is 65 years or older. The discount ($X = 1$) can only be given if the amount of the order is \$20.00 or over.

rule A	a student	0
	not a student	1
rule B	under 65 years old	0
	65 years or older	1
rule C	under \$20.00	0
	\$20.00 or over	1

- (i) Write the logic expression for this problem.

..... [2]

- (ii) Draw a logic circuit for this logic expression.

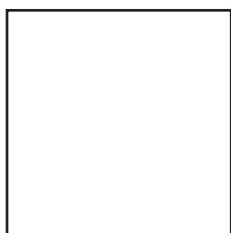
Each logic gate must have a maximum of **two** inputs.
Do **not** simplify this logic expression.



[3]

- (d) Draw the NAND logic gate symbol **and** complete the truth table for it.

NAND logic gate symbol:



Truth table:

A	B	X
0	0	
0	1	
1	0	
1	1	

[2]

